# Temperature Measurement

# Above

# 1 K

- Bodily sensation (hot vs cold) – not helpful
- Temperature is a property of matter. Temperature is one of the principal parameters of thermodynamics. On the microscopic scale, temperature is defined as the average energy of microscopic motions of a single particle in the system per degree of freedom. On the macroscopic scale, temperature is the unique physical property that determines the direction of heat flow between two objects placed in thermal contact.

- Internal energy, U, and entropy, $\sigma$, are the fundamental parameters describing any system. The temperature is DEFINED as

$$\frac{1}{\tau} = \left( \frac{\partial \sigma}{\partial U} \right)_{N.V}$$

the conventional temperature $\tau = k_B T$

and entropy $S = k_B \sigma$

# Early temperature scales

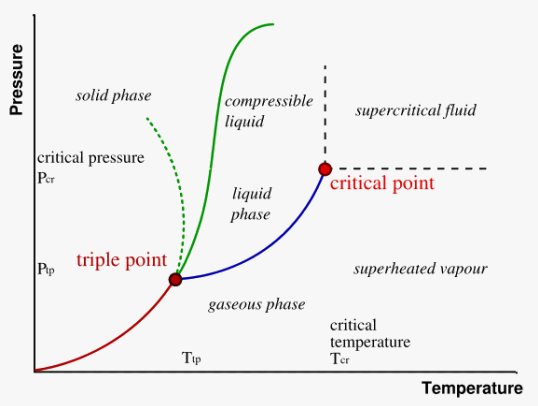
- In 1724 Daniel Gabriel Fahrenheit produced a temperature scale which now (slightly adjusted) bears his name. He could do this because he manufactured thermometers, using mercury (which has a high coefficient of expansion) for the first time and the quality of his production could provide a finer scale and greater reproducibility, leading to its general adoption.

- In 1742 Anders Celsius proposed a scale with zero at the boiling point and 100 degrees at the melting point of water, though the scale which now bears his name has them the other way around.

# The absolute scale

The **kelvin** (symbol: **K**) is a unit increment of temperature and is one of the seven SI base units. The **Kelvin scale** is a thermodynamic (absolute) temperature scale where absolute zero, the theoretical absence of all thermal energy, is zero (0 K). The Kelvin scale and the kelvin are named after the British physicist and engineer William Thomson, 1st Baron Kelvin (1824 - 1907), who wrote of the need for an “absolute thermometric scale”.

The kelvin unit and its scale, by international agreement, are defined by two points: absolute zero, and the triple point of Vienna Standard Mean Ocean Water




The single combination of pressure and temperature at which water, ice, and water vapor can coexist in a stable equilibrium occurs at exactly 273.16 K (0.01°C) and a partial vapor pressure of 611.73 pascals (6.1173 millibars, 0.0060373057 atm).

# INTRODUCTION

Determination of a temperature T with :

- Thermometer = device with reproducible f(T)
- Primary thermometer = f(T) is predictable
- Secondary thermometer =

f(T) needs to be calibrated with primary thermometers

# Primary Thermometers

A reasonable question at this point is: "What is a primary thermometer?" The answer: it is a thermometer that is based on a well understood physical principle and is reliable. In other words, it gives us the absolute temperature and gives repeatable results.

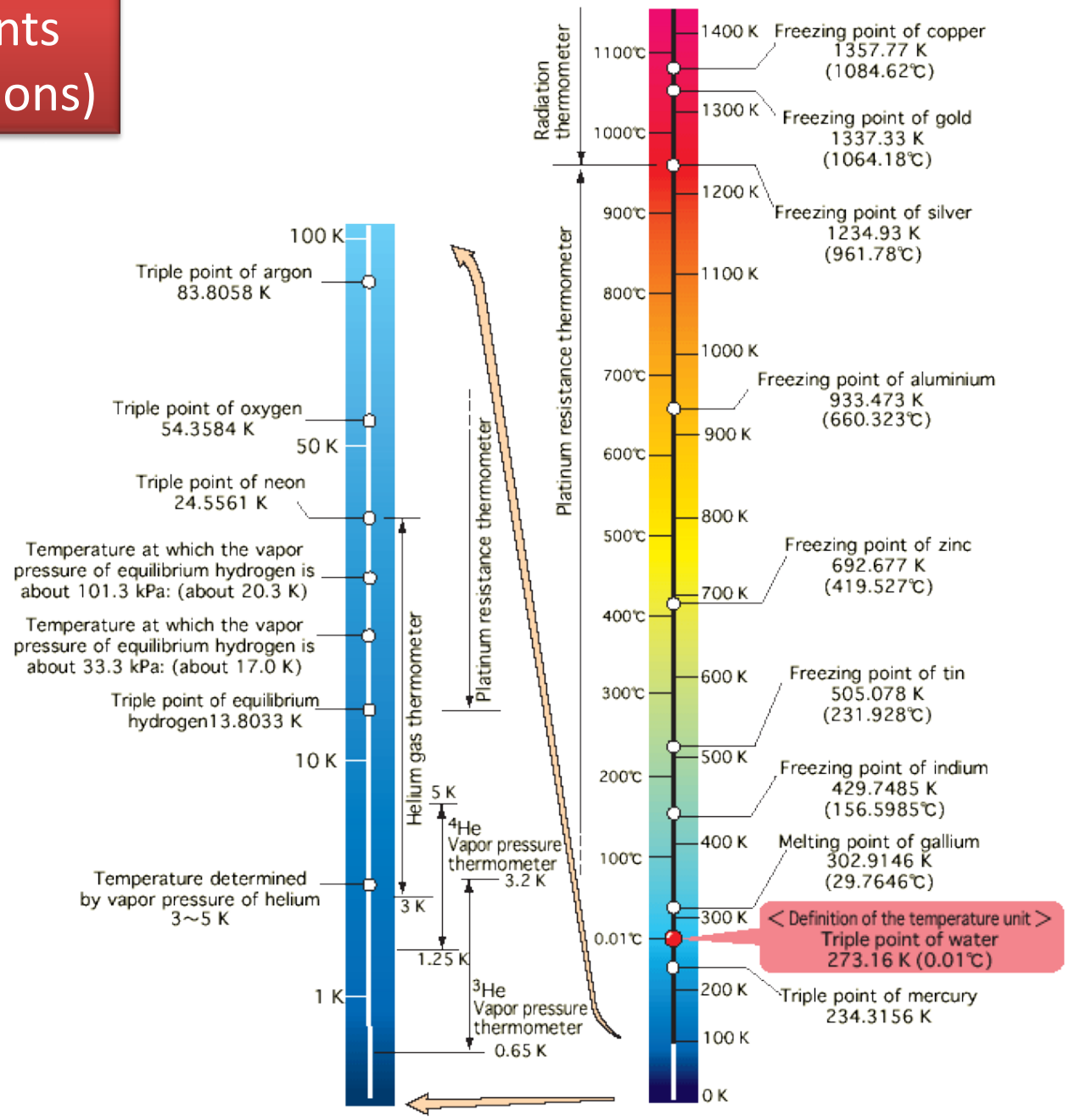
Several temperature scales for the superfluid $^3$He regime have been proposed; four of major interest are:

- The Helsinki scale: based on nuclear orientation thermometry.
- The La Jolla scale: based on noise thermometry.
- The Cornell scale: based on $^3$He melting curve thermometry.
- The Greywall scale: based on the properties of liquid $^3$He.

The four principal methods of primary thermometry are:

1. Gas thermometry in which the pressure is measured, based on the equation of state for an ideal gas, $PV = nRT$ (where $n$ is the number of moles of gas) with small corrections for non-ideality in a real gas such as helium.
2. Acoustic gas thermometry dependent on measuring the sound velocity in a gas, e.g. helium, and subject to the same virial corrections as gas thermometry (dielectric constant gas thermometry is similar).
3. Electrical noise in a resistor of $R$ ohms using the mean square voltage $\overline{V^2} = 4kTR\Delta f$ where $\Delta f$ is the frequency bandwidth.
4. Total black body radiation measurement.

# “Primary” reference points (do NOT rely on calibrations)



In practice:

Primary thermometer are (non exhaustive) : Gas, vapor or noise thermometers

Impractical for daily uses

Secondary thermometer are: of various types : resistance, diode ...

to be carefully chosen for a given purpose

# PRIMARY THERMOMETERS

## Gas thermometers : PV=NRT

N = known gas quantity V = known volume
P = measure
⇒ determination of T (tricky and only ok while some gas is left ... above low T

## Vapor thermometers : vapor/liquid coexistence

P = measure when the equilibrium is reached
⇒ determination of T
liquid/vapor used : 3He, 4He, N2, O2
magnetic field independent

## Magnetic thermometers : M = CH/T

(for paramagnetic substance)
C = Curie constant
H = applied magnetic field
M = measure
⇒ determination of T

# QUALITY OF MEASUREMENT: TEMPERATURE RANGE, RESOLUTION, REPRODUCIBILITY

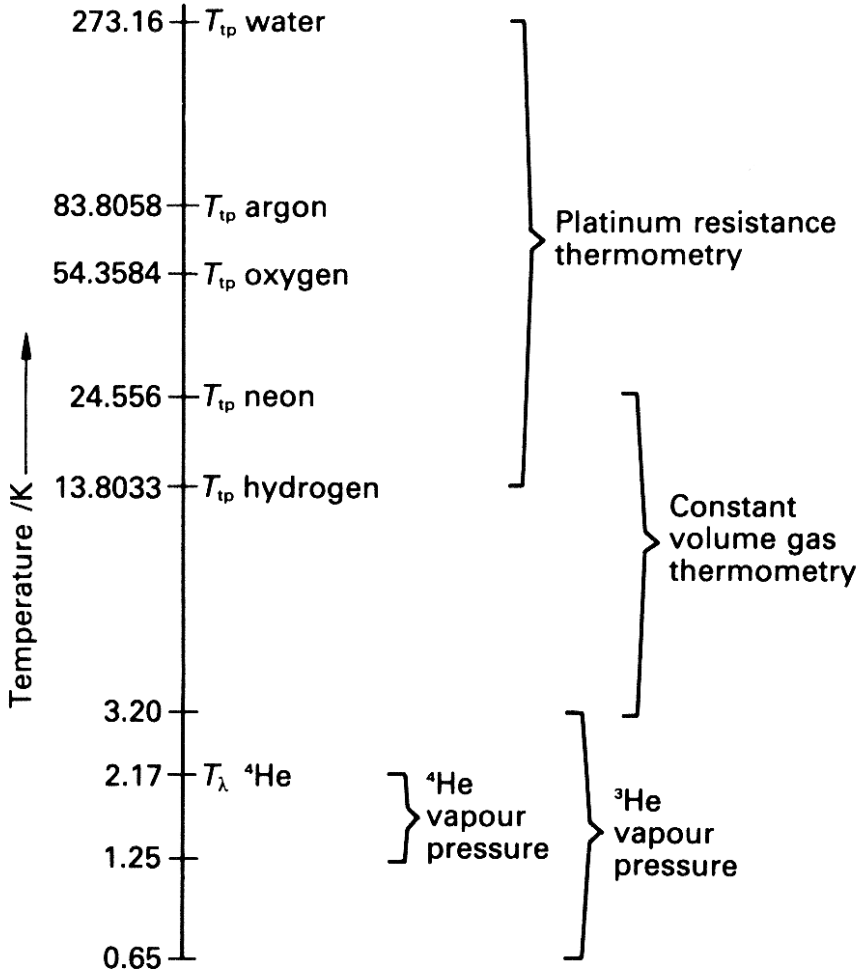
Secondary thermometers : sensor choice ?

Finance and utility :
cost and interchangeability (standard curve)

Environment :
magnetic field, vibration, ultra high vacuum

Experimental constraints :
size, power dissipation, thermal response time

# Different temperature ranges – different thermometers



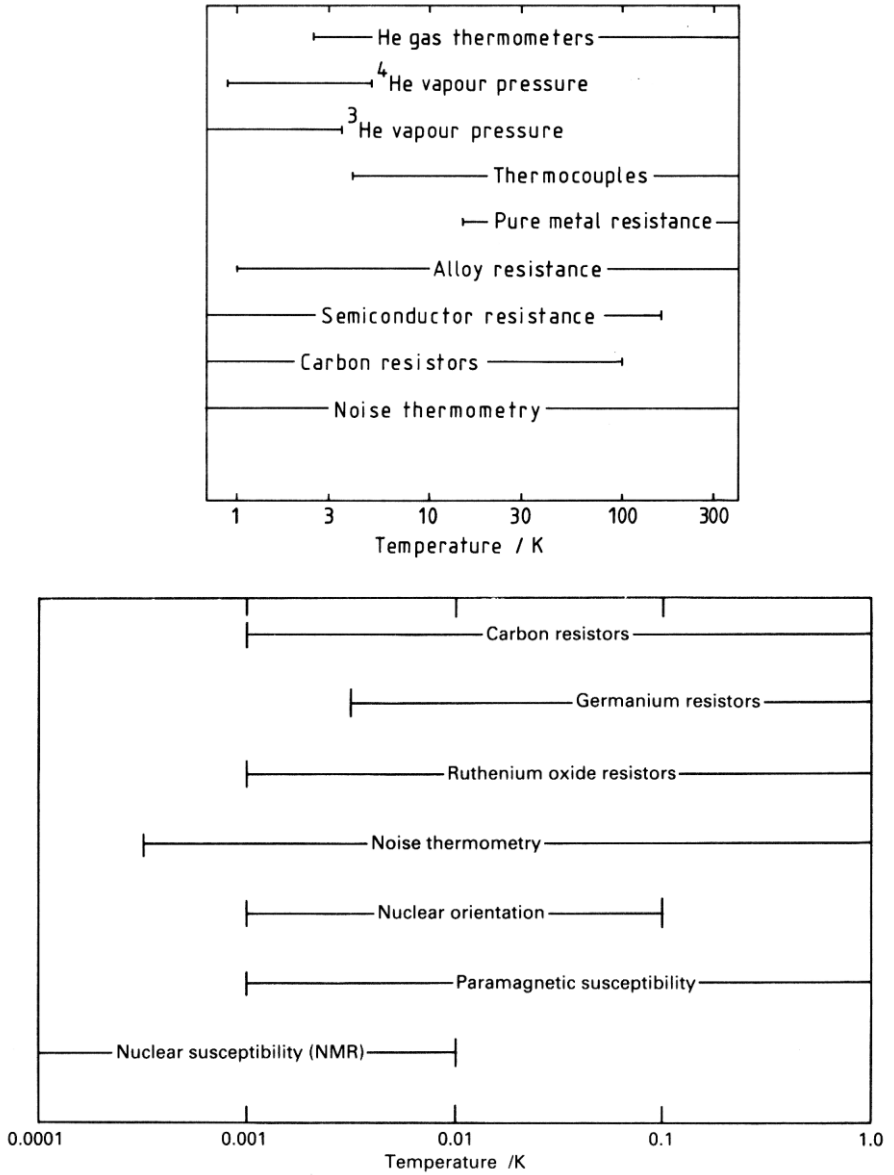



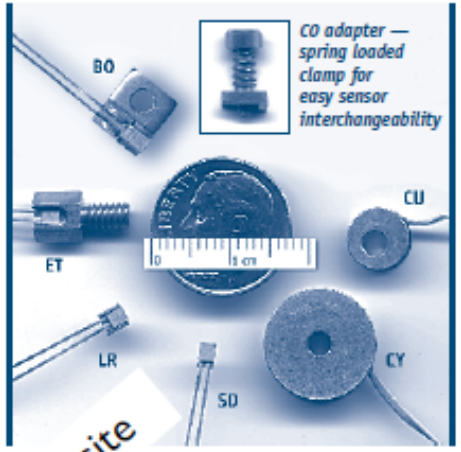

**Figure 7.11** *Types of thermometer suitable for use below 1 K, indicating the approximate temperature range over which they are used*

# Cryogenic Temperature Sensors

| | Temperature range | Standard curve | Below 1 K | Can be used in radiation | Performance in magnetic field |
|---|---|---|---|---|---|
| **Diodes** | | | | | |
| Silicon | 1.4 K to 500 K | x | | | Fair above 60 K |
| GaAlAs | 1.4 K to 500 K | | | | Fair |
| **Negative Temperature Coefficent RTDs** | | | | | |
| Cernox™ | 0.10 K to 325 K | | x | x | Excellent above 1 K |
| Germanium | 0.05 K to 100 K | | x | x | Not Recommended |
| Ruthenium Oxide (Rox™) | 0.01 K to 40 K | x | x | x | Good below 1 K |
| **Other** | | | | | |
| Thermocouples | 1.2 K to 1543 K | x | | | Fair |
| Capacitance | 1.4 K to 290 K | | | | Excellent |
| **Positive Temperature Coefficient RTDs** | | | | | |
| Platinum | 14 K to 873 K | x | | x | Fair above 30 K |
| Rhodium-Iron | 0.65 K to 500 K | | x | x | Fair above 77 K |

http://www.lakeshore.com/Products/Cryogenic-Temperature-Sensors/Pages/default.aspx

# DIODE




Principe : semiconductor, voltage is measured across a p-n junction.

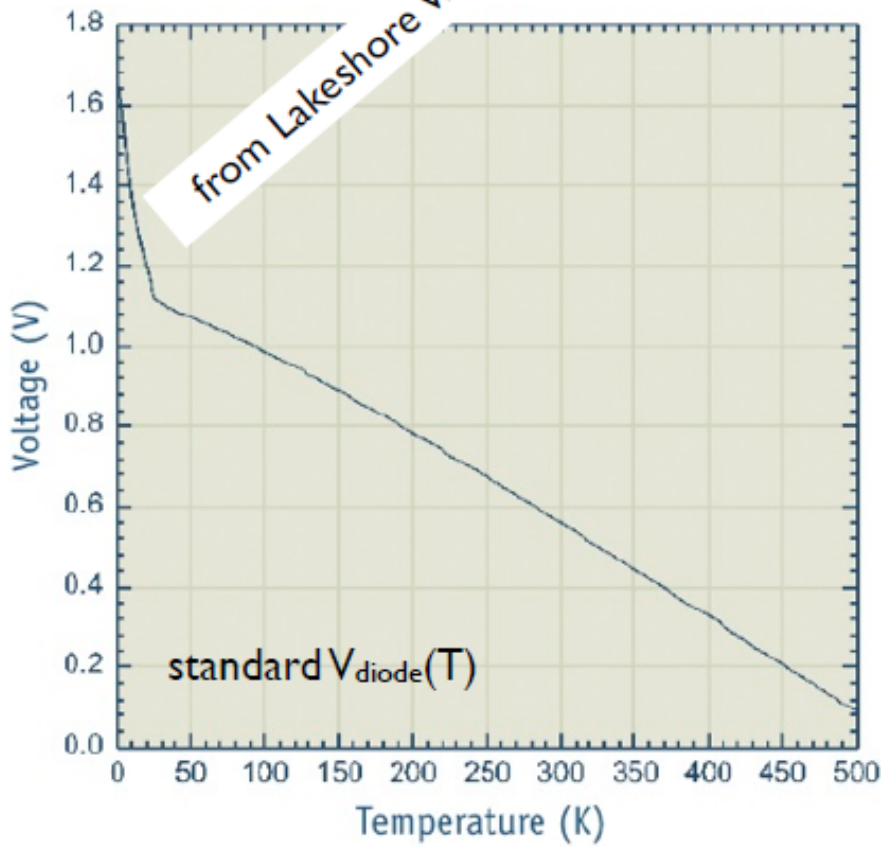
Most common type : SILICON

Temperature range : 1.4 K - 500 K

Interchangeability : yes
standard calibration within ±0.3 K

Cost : from 180$ (robust)

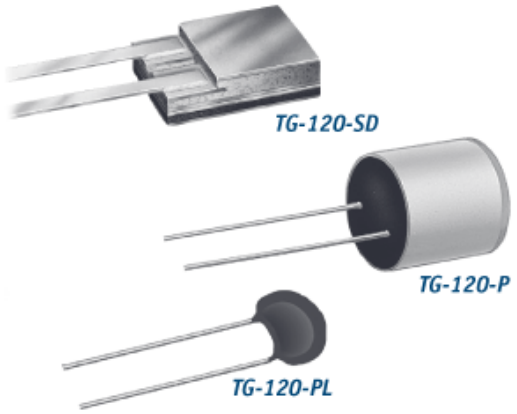
Magnetic Field :
sensitive, orientation issue (H//I is better)



# GaAlAs Diodes

## GaAlAs Diodes *

Product Overview | Tech Specs | Ordering Information | Downloads




### Description

The TG-120 gallium-aluminum-arsenide (GaAlAs) diode cryogenic temperature sensors are particularly well suited for low to moderate magnetic field applications at low temperatures. The GaAlAs sensing element exhibits high sensitivity (dV/dT) at low temperatures. Voltage-temperature characteristics are monotonic over the sensor's useful range from 1.4 K to 500 K (see data plots below).

Gallium-aluminum-arsenide diodes are direct band-gap, single junction devices that produce small output variances in the presence of magnetic fields. Consequently, their low magnetic field dependence makes them ideally suited for applications in moderate magnetic fields up to five tesla.

### TG-120-SD Features

- Monotonic temperature response from 1.4 K to 500 K
- Excellent sensitivity (dV/dT) at temperatures below 50 K
- Relatively low magnetic field-induced errors
- Rugged, reliable Lake Shore SD package designed to withstand repeated thermal cycling and minimize sensor self-heating
- Variety of packaging options

PACKAGING OPTIONS **P, PL, SD, CO, CU**

Click here for Adding Length to Sensor Leads

**The Lake Shore SD Package – The Most Rugged, Versatile Package in the Industry**

*The SD package, with direct sensor-to-sapphire base mounting, hermetic seal, and welded platinum leads, provides the industry's most rugged, versatile sensors with the best sample to chip connection. Designed so heat coming down the leads bypasses the chip, it can survive several thousand hours at 500 K (depending on model) and is compatible with most ultra high vacuum applications. It can be indium soldered to samples.*

### TG-120-P Features

- Temperature range: 1.4 K to 325 K
- Reproducibility at 4.2 K: ±10 mK

# Pt Resistance Thermometers

## PT-100 Series Platinum RTDs

Product Overview | Tech Specs | Ordering Information | Downloads



PT-102



PT-103



PT-111

## Description

PT-100 platinum resistance thermometers (PRTs) are an excellent choice for use as cryogenic temperature sensing and control elements in the range from 30 K to 873 K (-243 °C to 600 °C). Over this temperature span, PRTs offer high repeatability and nearly constant sensitivity (dR/dT). Platinum resistors are also useful as control elements in magnetic field environments where errors approaching one degree can be tolerated. PRTs are interchangeable above 70 K. The use of controlled-purity platinum assures uniformity from one device to another.

PRTs experience rapidly decreasing sensitivity below approximately 30 K. They should be calibrated in order to achieve maximum accuracy for use below 100 K. The plot illustrates platinum sensor conformance to the IEC 751 curve.

## Features

- Temperature range: 14 K to 873 K (model dependant)
- Conforms to IEC 751 standards down to 70 K
- High reproducibility: ±5 mK at 77 K
- Low magnetic field dependence above 40 K
- Excellent for use in ionizing radiation
- SoftCal™ calibration available

PACKAGING OPTIONS **AL, AM**

▸ Click here for Adding Length to Sensor Leads

### Matching

*If your application requires more than one platinum resistor, up to five platinum resistors can be matched to one another to within ±0.1 K at liquid nitrogen temperature with the purchase of only one calibration.*

# Rhodium-Iron RTDs

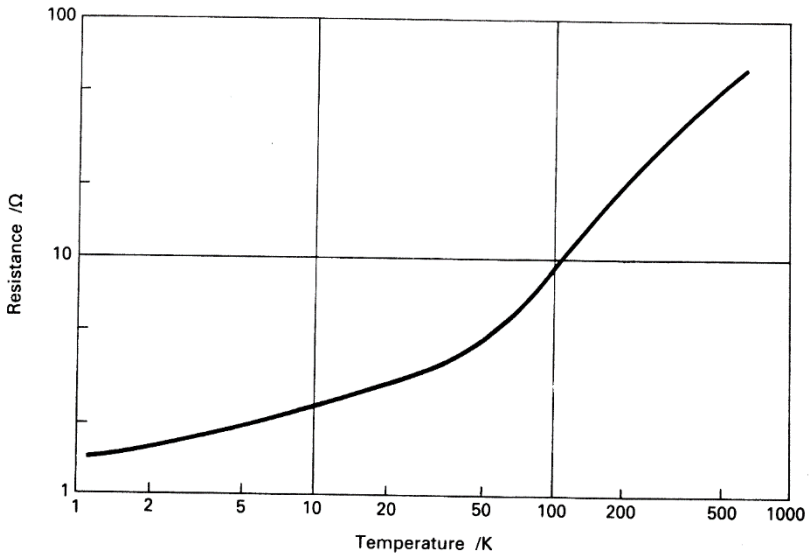



**Figure 7.8** *Resistance–temperature characteristics for a typical Rhodium–iron resistance thermometer [Courtesy of Oxford Instruments UK Ltd.]*

## Rhodium-Iron RTDs

Product Overview | Tech Specs | Ordering Information | Downloads







### Features

- Good long term stability: ±10 mK from 1.4 K to 325 K
- RF-800 offers a wide temperature range from 0.65 K to 500 K
- Linear response above 100 K
- Excellent resistance to ionizing radiation
- Small chip size available with extremely fast thermal response time

**The RF-100U and RF-100T sensors have been discontinued and there is only a limited quantity available. Please refer to http://www.lakeshore.com/eod/obsa.html for remaining quantities. Alternative sensors are Cernox for measuring temperatures from 100 mK to 420 K or Platinum for measuring temperatures from 14 K to 873 K.**

### Description

Rhodium-iron cryogenic temperature sensors offer a positive temperature coefficient, monotonic response over a wide temperature range, and high resistance to ionizing radiation.

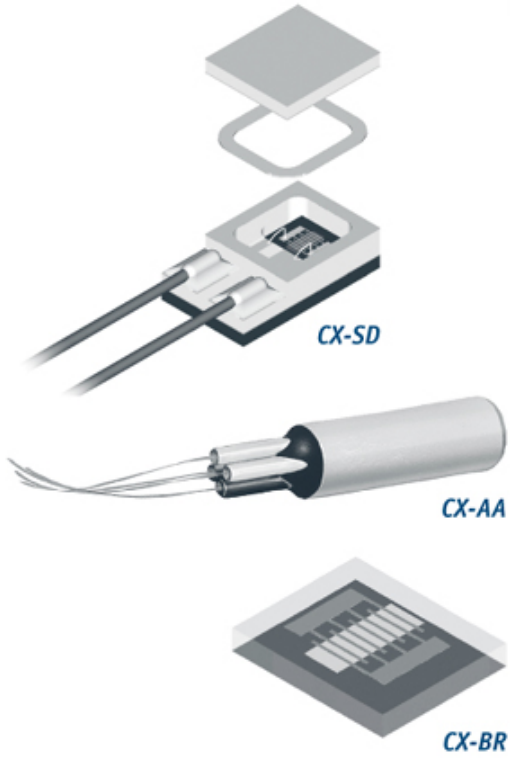
#### RF-100

The Lake Shore thin film rhodium-iron temperature sensor offers significant advantages over comparable wire-wound resistance sensors. The thin film sensors feature a smaller package size, which makes them useful in a broader range of experimental mounting schemes, and they are available at a much lower cost. Additionally, they have proven to be very stable over repeated thermal cycling and under extended exposure to ionizing radiation. Furthermore, the thermal time constant of thin film rhodium-iron cryogenic temperature sensors (bare chip) is on the order of milliseconds, while the thermal time constant of wire-wound resistors is on the order of seconds.

#### RF-800

The RF-800 rhodium-iron resistance sensor features monotonically decreasing resistivity from 500 K to 0.65 K, although sensitivity (dR/dT) falls off in the region of 30 K. From 100 K to 273 K the resistance changes linearly with temperature to within 1 K. RF-800-4 sensors also exhibit monotonic response at higher temperatures, hence their adaptability for use over the broad range from 1.4 K to 500 K.

# Negative T coefficient: Cernox




## Description

Cernox™ thin film resistance cryogenic temperature sensors offer significant advantages over comparable bulk or thick film resistance sensors. The smaller package size of these thin film sensors makes them useful in a broader range of experimental mounting schemes, and they are also available in a chip form. They are easily mounted in packages designed for excellent heat transfer, yielding a characteristic thermal response time much faster than possible with bulk devices requiring strain-free mounting. Additionally, they have been proven very stable over repeated thermal cycling and under extended exposure to ionizing radiation.

PACKAGING OPTIONS **AA, BC, BG, BO, BR, CD, CO, CU, ET, LR, MR, SD**

▸ Click here for Adding Length to Sensor Leads

**CX-1010 – the Ideal Replacement for Germanium RTDs**

*The CX-1010 is the first Cernox™ designed to operate down to 100 mK, making it an ideal replacement for Germanium RTDs. Unlike Germanium, all Cernox models have the added advantage of being able to be used to room temperature. In addition, Cernox is offered in the incredibly robust Lake Shore SD package, giving researchers more flexibility in sensor mounting.*

**The Lake Shore SD Package – The Most Rugged, Versatile Package in the Industry**

*The SD package, with direct sensor-to-sapphire base mounting, hermetic seal, and soldered copper leads, provides the industry's most rugged, versatile cryogenic temperature sensors with the best sample to chip connection. Designed so heat coming down the leads bypasses the chip, it can survive several thousand hours at 420 K (depending on model) and is compatible with most ultra high vacuum applications. It can be indium soldered to samples without sensor calibration shift.*

## Features

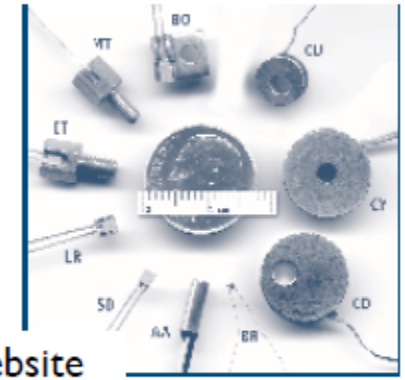
- Low magnetic field- induced errors
- Temperature range of 100 mK to 420 K (model dependent)
- High sensitivity at low temperatures and good sensitivity over a broad range
- Excellent resistance to ionizing radiation
- Bare die cryogenic temperature sensor with fast characteristic thermal response times: 1.5 ms at 4.2 K, 50 ms at 77 K
- Broad selection of models to meet your thermometry needs
- Excellent stability
- Variety of packaging options

# RESISTANCE : negative T coefficient

Principe :

semiconductors, R(T) ↑as T ↓.

Type : CERNOX (replacement of Carbon-glass)




from Lakeshore website

Temperature range :

0.1 K - 420 K owing doping level

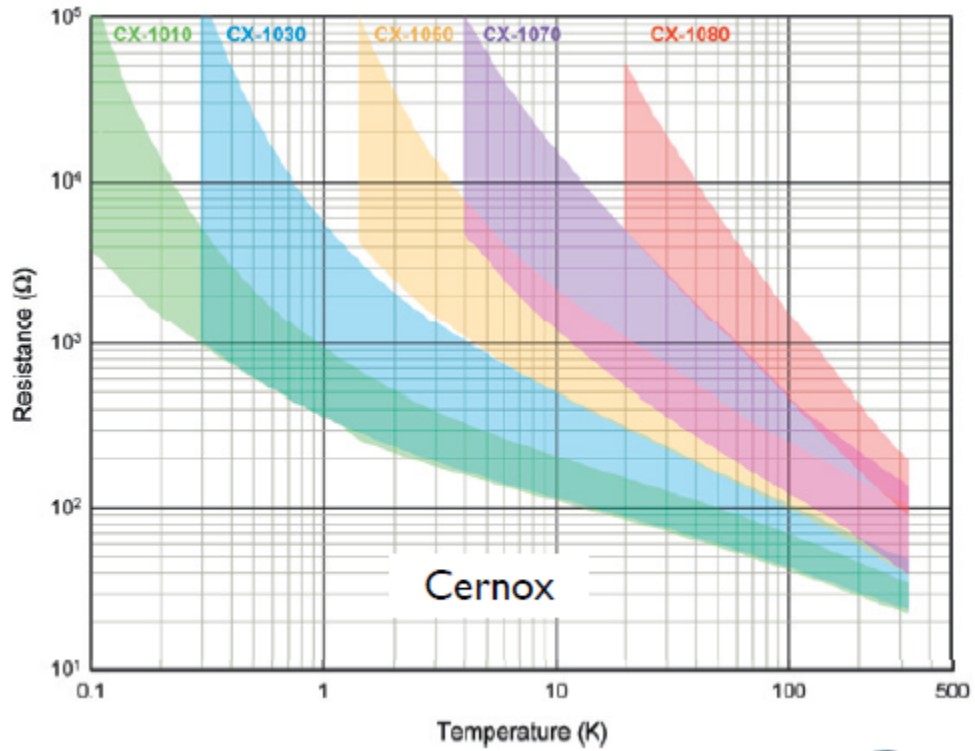
Interchangeability : NO

Cost : from 100$ (quite robust)

Magnetic Field :

$|\Delta T| < 0.4$ K for T>4K and H<8T

→ useful under field



# Germanium Resistance

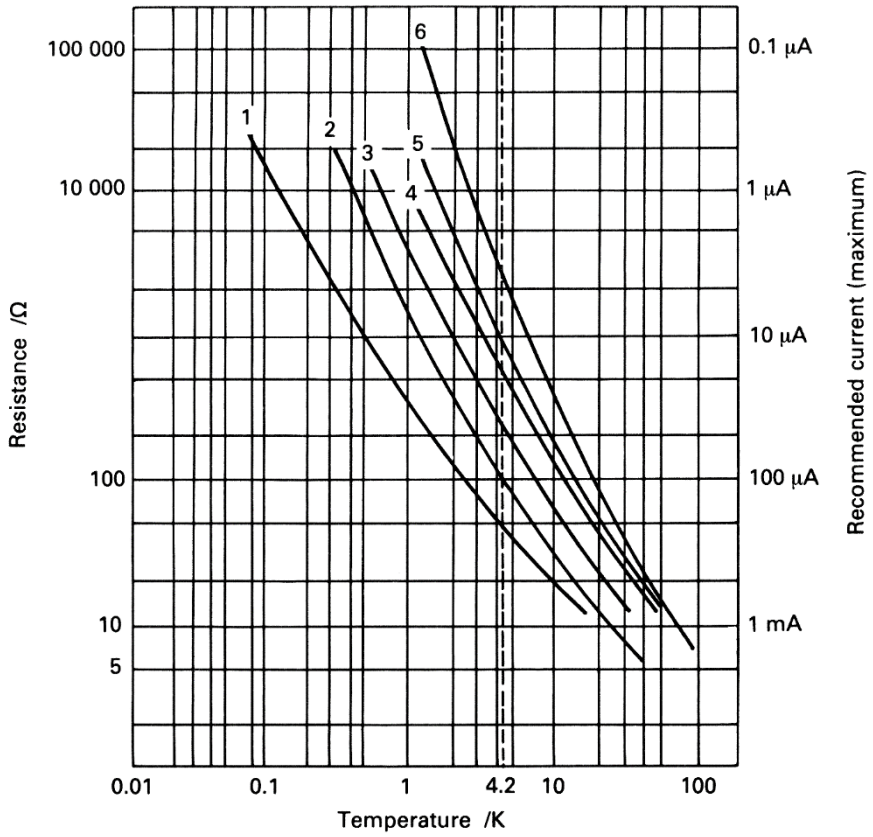
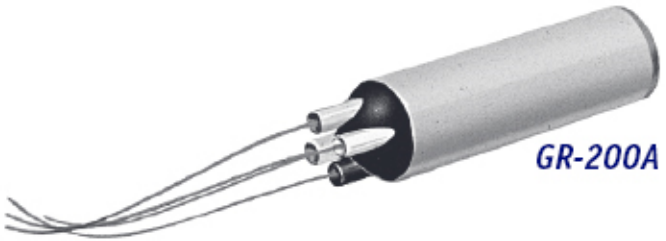



**Figure 7.9** *Typical resistance–temperature characteristics for a selection of germanium thermometer elements. The thermometers are characterised by their resistance at 4.2 K. That is, 50, 100, 250, 500, 1000 and 1500 Ω for curves 1–6 respectively. Also shown is the maximum recommended measurement current [Courtesy of Oxford Instruments UK Ltd.]*

## Germanium RTDs

Product Overview | Tech Specs | Ordering Information | Downloads







### Description

The GR-200 Germanium Resistance Temperature Sensor is recognized as a "Secondary Standard Thermometer" and has been employed in the measurement of temperature from 0.05 K to 30 K for more than 30 years.

GR-200 sensors have a useful temperature range of about two orders of magnitude. The exact range depends upon the doping of the germanium element. Cryogenic temperature sensors with ranges from below 0.05 K to 100 K are available. Between 100 K and 300 K, dR/dT changes sign and dR/dT above 100 K is very small for all models. Sensor resistance varies from several ohms at its upper useful temperature to several tens of kilohms at its lower temperature.

### Features

- Recognized as a "Secondary Standard Thermometer"
- High sensitivity provides submillikelvin control at 4.2 K and below
- Excellent reproducibility better than ±0.5 mK at 4.2 K
- Various models for use from 0.05 K to 100 K
- Excellent resistance to ionizing radiation

Because device sensitivity increases rapidly with decreasing temperature, a high degree of resolution is achieved at lower temperatures, making these resistors very useful for submillikelvin control at 4.2 K and below.

The GR-200 sensors have excellent stability, and ±0.5 mK reproducibility at 4.2 K. The germanium resistor is usually the best choice for high-accuracy work below 30 K. Use in a magnetic field is not recommended.

# RESISTANCE : negative T coefficient

Principe :
semiconductors, R(T) ↑as T ↓.

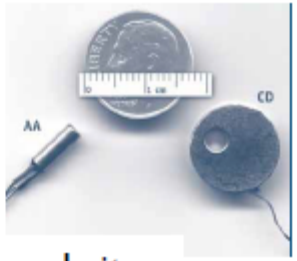
Type : Germanium (Ge)

Temperature range : 0.05 K - 100 K
very accurate and sensitive.

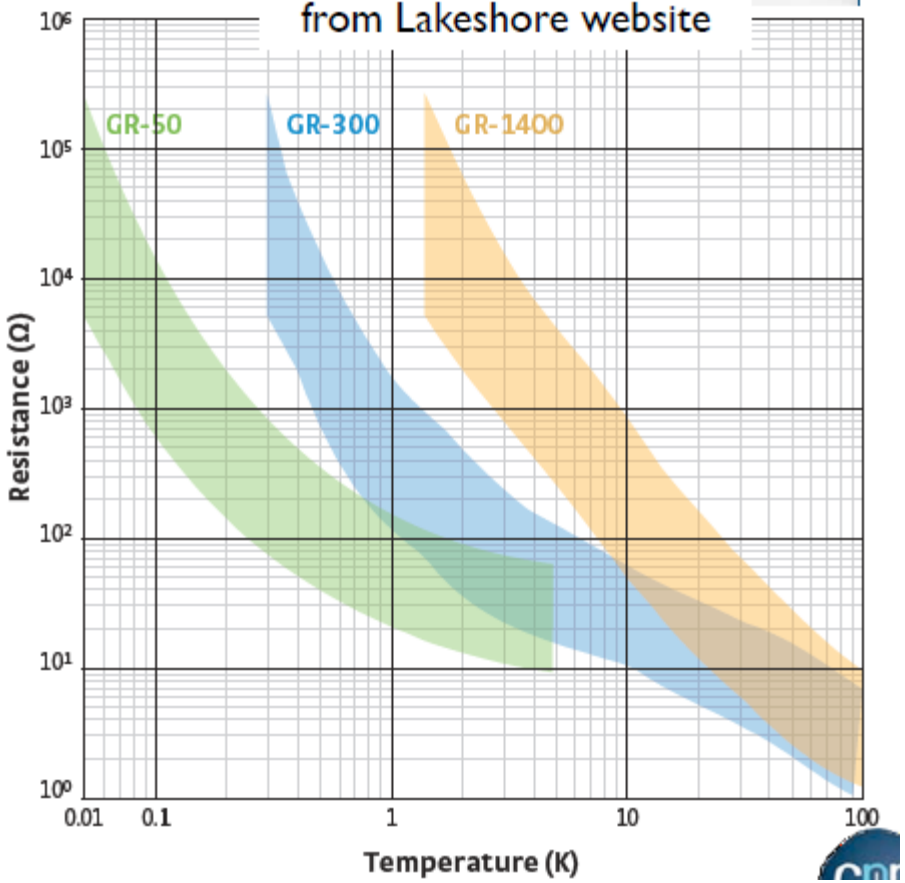
Interchangeability : no

Cost : from 286$ (fragile)

Magnetic Field : sensitive




from Lakeshore website



# Carbon-Glass



CGR

## Description

Carbon-Glass RTDs (CGRs) have the longest history of use of any cryogenic temperature sensor suitable for high magnetic fields and wide range temperature sensing. These resistance cryogenic temperature sensors are highly reproducible and can be used from 1.4 K to 100 K and in magnetic fields up to 20 tesla. Their extremely high sensitivity at liquid helium temperatures makes them very useful for submillikelvin control below 10 K. CGR sensors are monotonic in resistance temperature characteristic between 1.4 K and 325 K, but their reduced sensitivity (≈0.01 W/K) above 100 K limits their usage at higher temperatures.

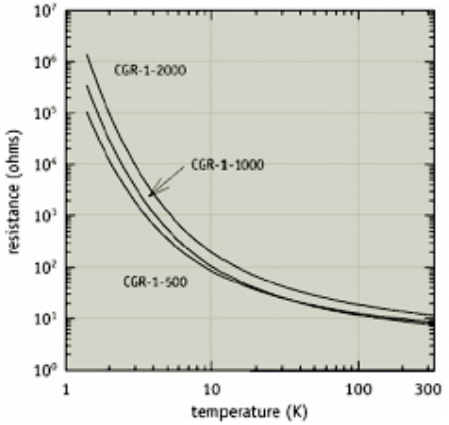
## Features

- Low magnetic field induced errors
- For use in magnetic fields up to 20 tesla
- Reproducible in the 1.4 K to 100 K range
- Monotonic R vs. T and dR/dT vs. T response curves
- High sensitivity provides submillikelvin control at 4.2 K and below
- Usable sensitivity over the broad range of 1.4 K to 325 K
- Good resistance to ionizing radiation at low temperatures

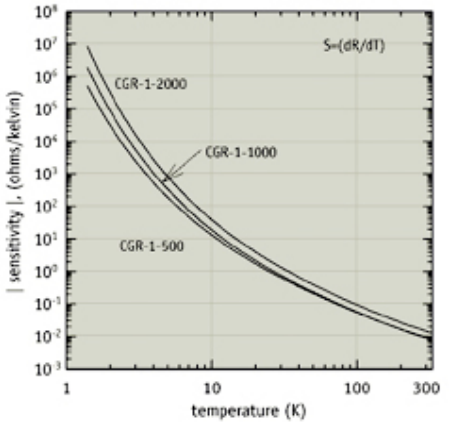
PACKAGING OPTIONS **AA, CD**

▸ Click here for Adding Length to Sensor Leads

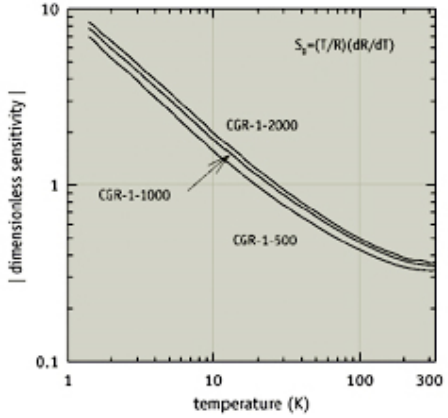
### Typical Carbon-Glass Resistance Values




### Typical Carbon-Glass Sensitivity Values




### Typical Carbon-Glass Dimensionless Sensitivity Values



# Ruthenium Oxide




## Features

### RX-102A Features

- Standard curve interchangeable
- Good radiation resistance
- Useful down to 50 mK
- Low magnetic fieldinduced errors

### RX-102B Features

- Useful down to 10 mK; calibrations down to 20 mK available
- Monotonic from 10 mK to 300 K

### RX-202A Features

- Standard curve interchangeable
- Good radiation resistance
- Monotonic from 50 mK to 300 K
- 4× improvement in magnetic field-induced errors over other ruthenium oxides

### RX-103A Features

- Standard curve interchangeable
- Good radiation resistance
- Best choice for interchangeability from 1.4 K to 40 K
- Low magnetic fieldinduced errors

## Description

Ruthenium oxide temperature sensors are thick-film resistors used in applications involving magnetic fields. These composite sensors consist of bismuth ruthenate, ruthenium oxides, binders, and other compounds that allow them to obtain the necessary temperature and resistance characteristics. Each Lake Shore Rox™ model adheres to a single resistance versus temperature curve.

### RX-102A

The RX-102A (1000 Ω at room temperature) is useful down to 50 mK and has better interchangeability than the RX-202A as well as low magnetic field induced errors below 1 K.

### RX-102B-CB

The RX-102B-CB (1000 Ω at room temperature) is useful down to 10 mK (calibrations available down to 20 mK) and monotonic from 10 mK to 300 K. The unique package design maximizes thermal connection and minimizes heat capacity at ultra low temperatures. The RX-102BCB is not interchangeable to a standard curve and not recommended for use in magnetic fields.

### RX-202A

The RX-202A (2000 Ω at room temperature) has a 4× improvement in magnetic field-induced errors over other commercially available ruthenium oxide temperature sensors with similar resistances and sensitivities. Most ruthenium oxide sensors have a maximum useful temperature limit well below room temperature, where the sensitivity changes from negative to positive. The RX-202A however, is designed to have a monotonic response from 0.05 K up to 300 K.

### RX-103A

The RX-103A (10,000 Ω at room temperature) has a unique resistance and temperature response curve combined with low magnetic fieldinduced errors, and is the best choice for interchangeability from 1.4 K to 40 K.

PACKAGING OPTIONS **AA, CB, BR, CD**

Click here for Adding Length to Sensor Leads

# RESISTANCE : negative T coefficient

Principe :
semiconductors, R(T) ↑as T ↓.

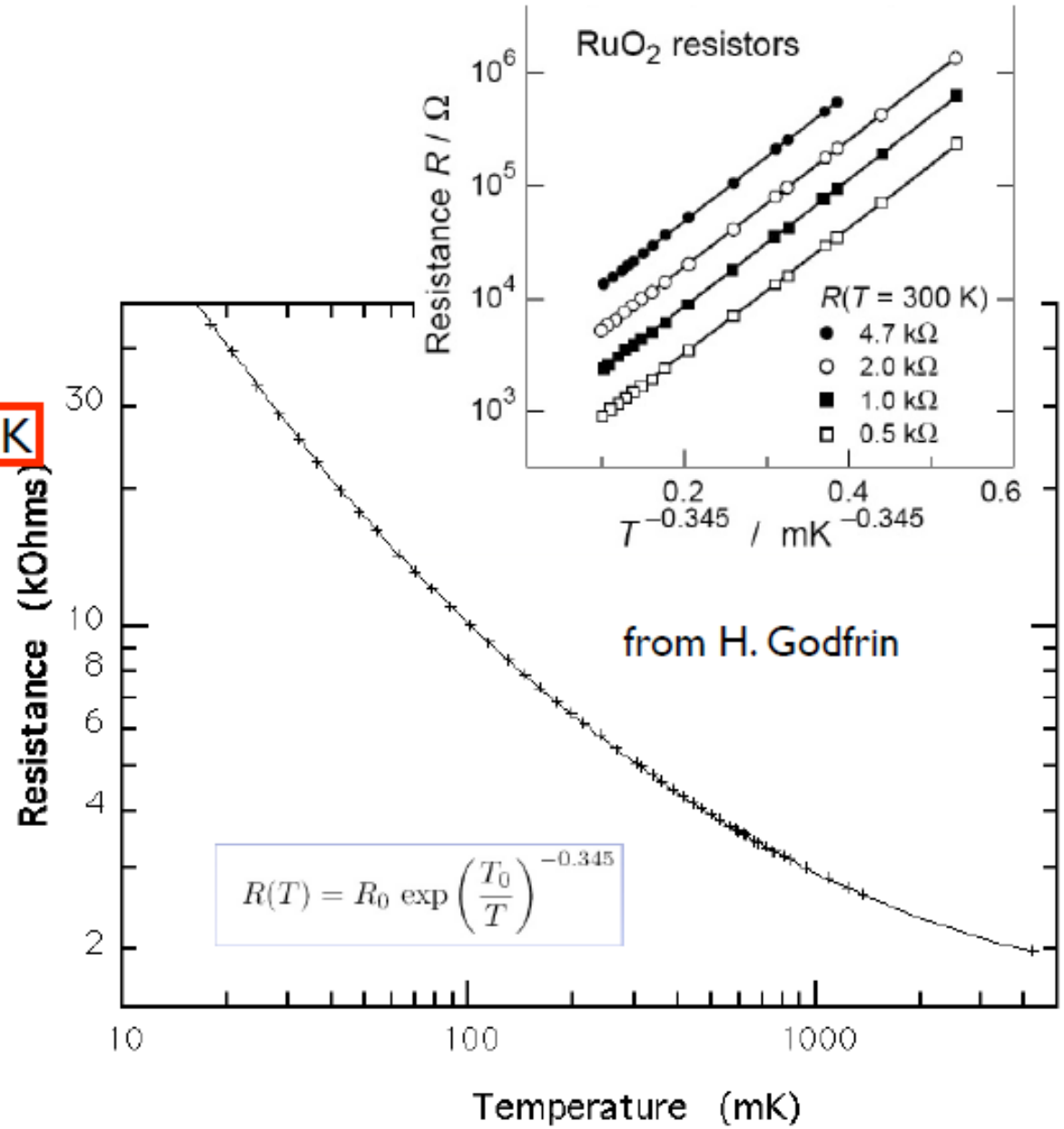
Type : Ruthenium Oxide ($RuO_2$)

Temperature range : 0.01 K - 40 K

Interchangeability :
yes but error up to 5K

Cost : from 99$

Magnetic Field :
useful below 1K but sensitive



# THERMOCOUPLE

- Principe : Seebeck effect, V=SxΔT
- junction of  positive and negative thermoelements

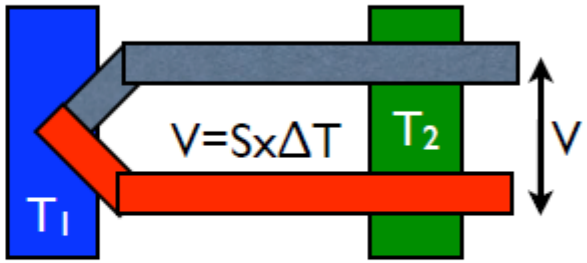
- Example type : «Chromel-Gold/Iron 0.07%»
  - negative thermoelt = Au + 0.07% Fe
  - positive thermoelt = Ni-Cr alloy

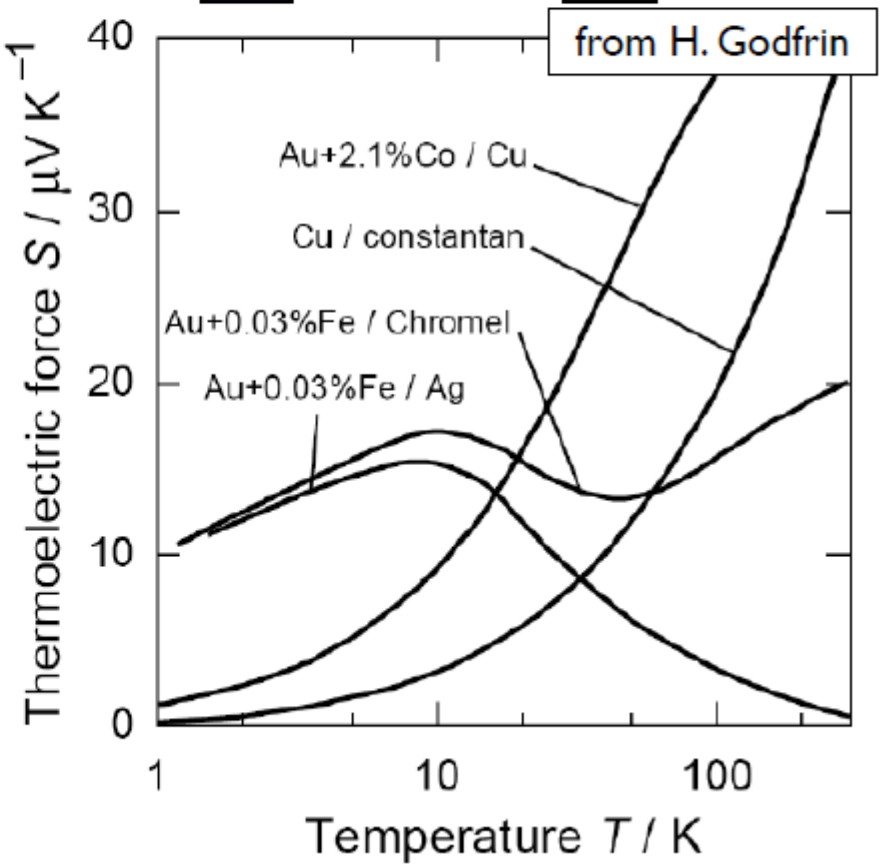
- Temperature range : 1.2 K - 1500 K

- Interchangeability : yes

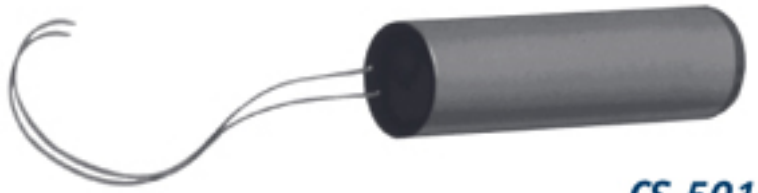
- Cost : from 50$,
- small, robust, easy to measure

- Magnetic Field : sensitive see lakeshore data



# capacitance



CS-501GR

## Features

- Virtually no magnetic field-induced errors
- Capable of mK control stability in the presence of strong magnetic fields
- Monotonic in C versus T to nearly room temperature

## Description

CS capacitance sensors are ideally suited for use as cryogenic temperature control sensors in strong magnetic fields because they exhibit virtually no magnetic field dependence. Displacement current is not affected by magnetic fields. Consequently, temperature control fluctuations are kept to a minimum when sweeping magnetic field or when changing field values under constant temperature operation.

Because small variations in the capacitance/temperature curves occur upon thermal cycling, calibrations must be transferred to the capacitor from another sensor after cooling for the best accuracy. It is recommended that temperature in zero field be measured with another temperature sensor and that the capacitance sensor be employed as a control element only.

# CAPACITANCE




Principe : C(T),
NO variation with field B,
!!! variation with thermal cycling.

Purpose :
precise regulation under field B
in complement to a thermometer

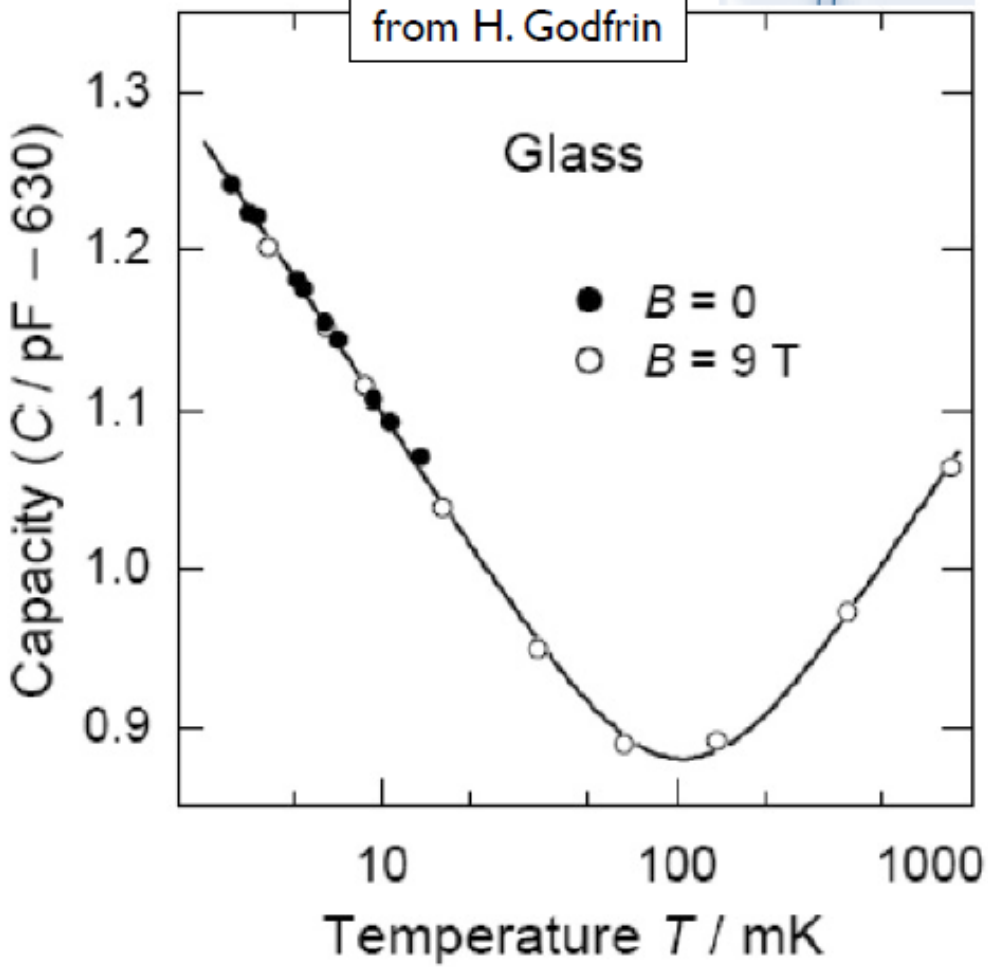
Temperature range : 1.4 K - 290 K

Interchangeability : no

Cost : from 440$

from Lakeshore website

from H. Godfrin



# MEASUREMENT TECHNICS

Diode set-up

Resistance set-up

General requirements (including wiring precautions)

General Wheatstone bridge technique

# DIODE SET- UP

from Lakeshore website

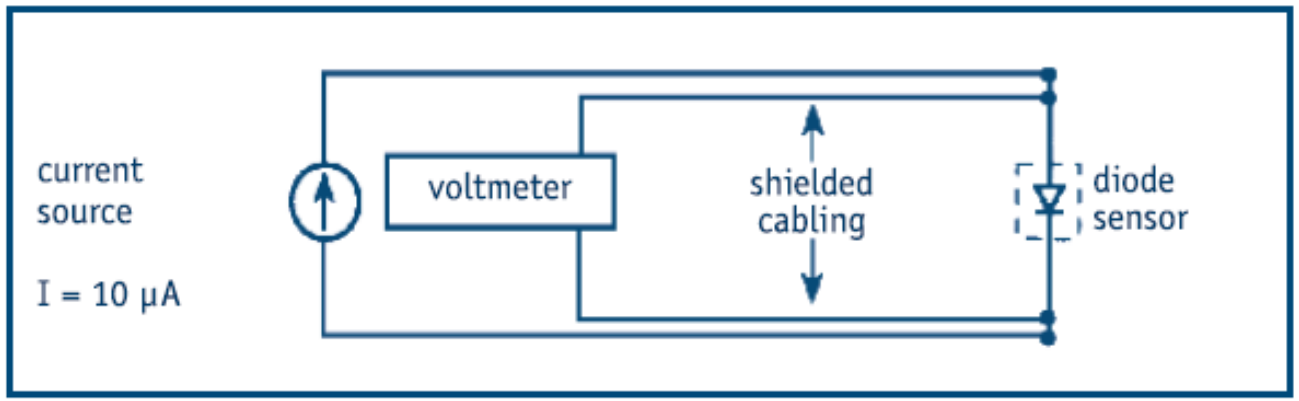



***Figure 2** – Typical diode sensor instrumentation schematic*

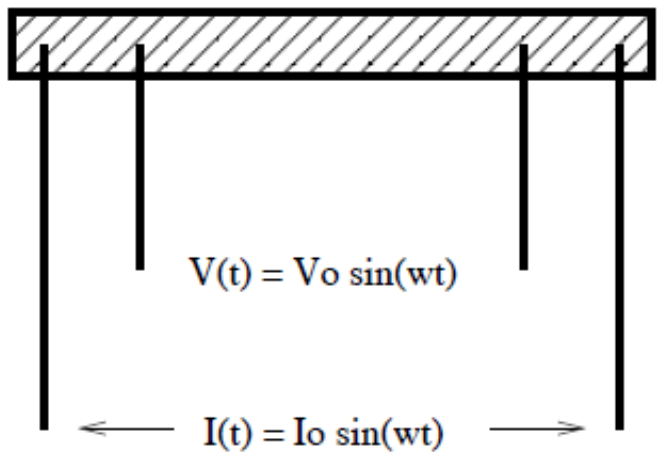
I = 10µA → V(300 K) = 0.5 V and V(77 K) = 1V

! Wiring ! : noise currents produce shift in measurement
Solution : AC-filters

# RESISTANCE SET- UP

4 LEADS TECHNIQUE (with a lock-in)

Input : AC current I → Output : AC voltage V




Determination of R by V=RI
TO GET RID OF THE RESISTANCE OF THE WIRES

When R decreases linearly with T : single current excitation ~ 1mA (ex : Pt)
When R strongly ↑ as T ↓ : current must be varied from ~0.01µA to ~ 1mA (ex: cernox)

# MEASUREMENT TECHNICS

**General requirements :**

- **measurement set-up impedance** : Zset-up>>Zthermometer

- **grounding** : ONE reference for all instruments and cryostat

→ to avoid current flows and thus voltage offsets

- **reduce AC signal interference :** problem : wires = antennas

  i/ **continuous conductive shielding** connect to the ground from the cryostat to the instruments

  ii/ **twisted and short wires pairs** (= bad antenna)

  iii/ if needed additional AC-filters

# MEASUREMENT TECHNICS

**General requirements :**

- wiring thermalization to avoid heat load into the thermometer

- thermal contact of the thermometer : screw, clamp, GE varnish ...

- thermometer self-heating ? (test/adapt the current value)

- configuration between thermometer and sample : good thermal contact ?

- time constant of the sensor (sensor construction + thermal contact and distance from the heater ...)

# MEASUREMENT TECHNICS

**The Wheatstone bridge :**

to measure small changes of resistance (capacitance or inductance)

R — three known resistances, **$R_3$ can be adjusted**

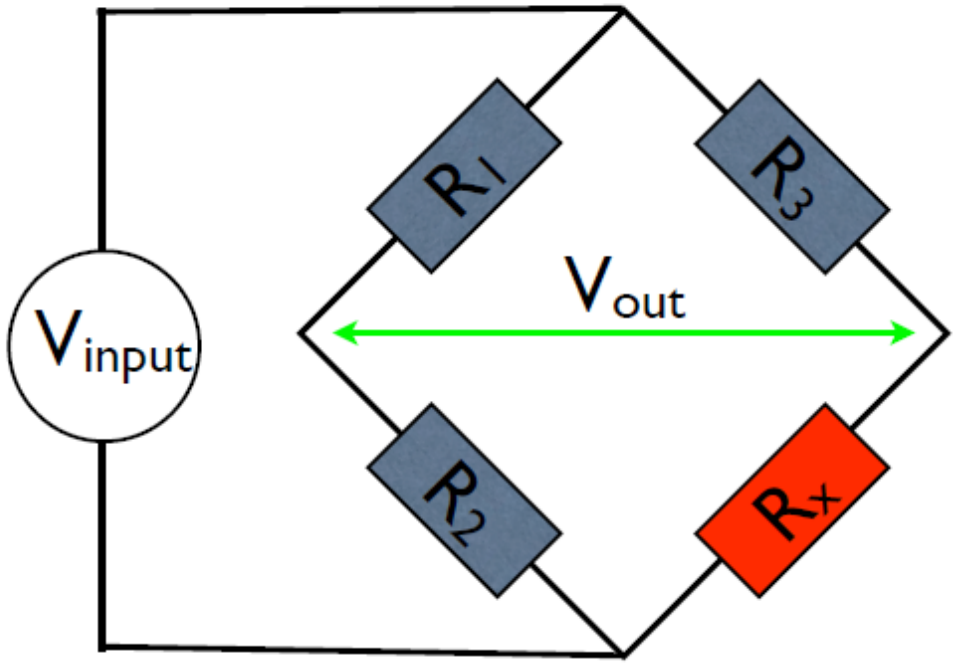
$R_x$ — sensor resistance : to be determined

**Bridge balance**

**when $V_{out} = 0$**

**then**

**$R_x = R_2 . R_3 / R_1$**




Cryogenic resistance bridges are based on the Wheastone bridge principle.

# REGULATION ISSUE

How to adjust the heat Q to regulate a temperature ?

PID technique : commonly used

P = proportional, Q = P. (Tset-Tmeas)

I = integral, Q = I . ∑ (Tset-Tmeas)

D = derivative, Q = D. d (Tset-Tmeas)/dt

P ~ on/off regulation

I ~ smooth the P regulation (if I is not too high, otherwise T wobbles)

D ~ fasten the regulation IF the measurement is not too noisy

# REGULATION ISSUE

How to REALLY regulate a temperature under a magnetic field ?

**Regulation on a capacitance**

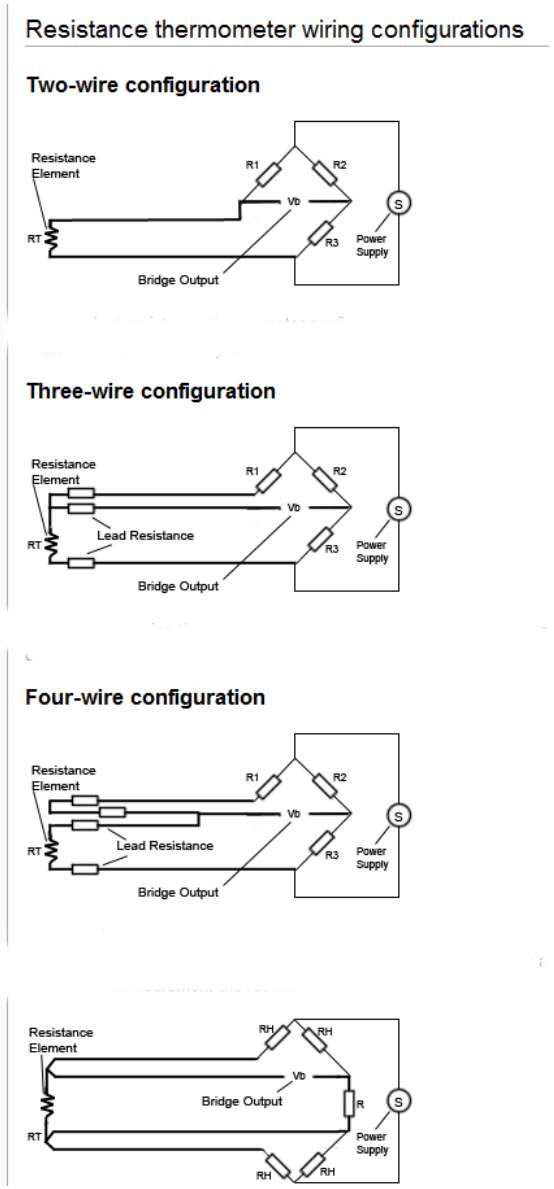
(or on an insensitive sensor but very few other type are available)

1°/ H = 0 T : regulation on a sensor R

-> determination of the capacitance value at T : C

2°/ H-sweep : regulation to maintain C constant (bridge technique)

# Resistance thermometers




- Resistance thermometers, also called resistance temperature detectors (RTDs), are temperature sensors that exploit the predictable change in electrical resistance of some materials with changing temperature.

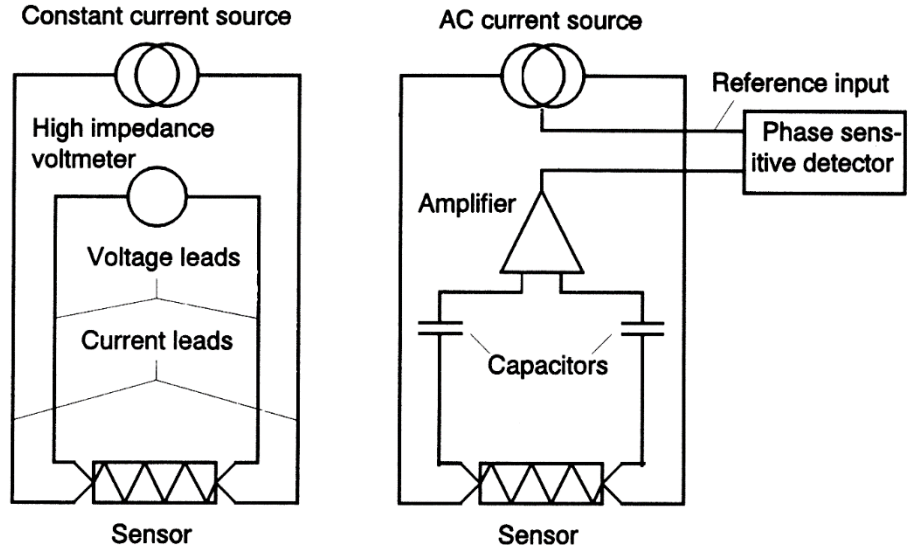
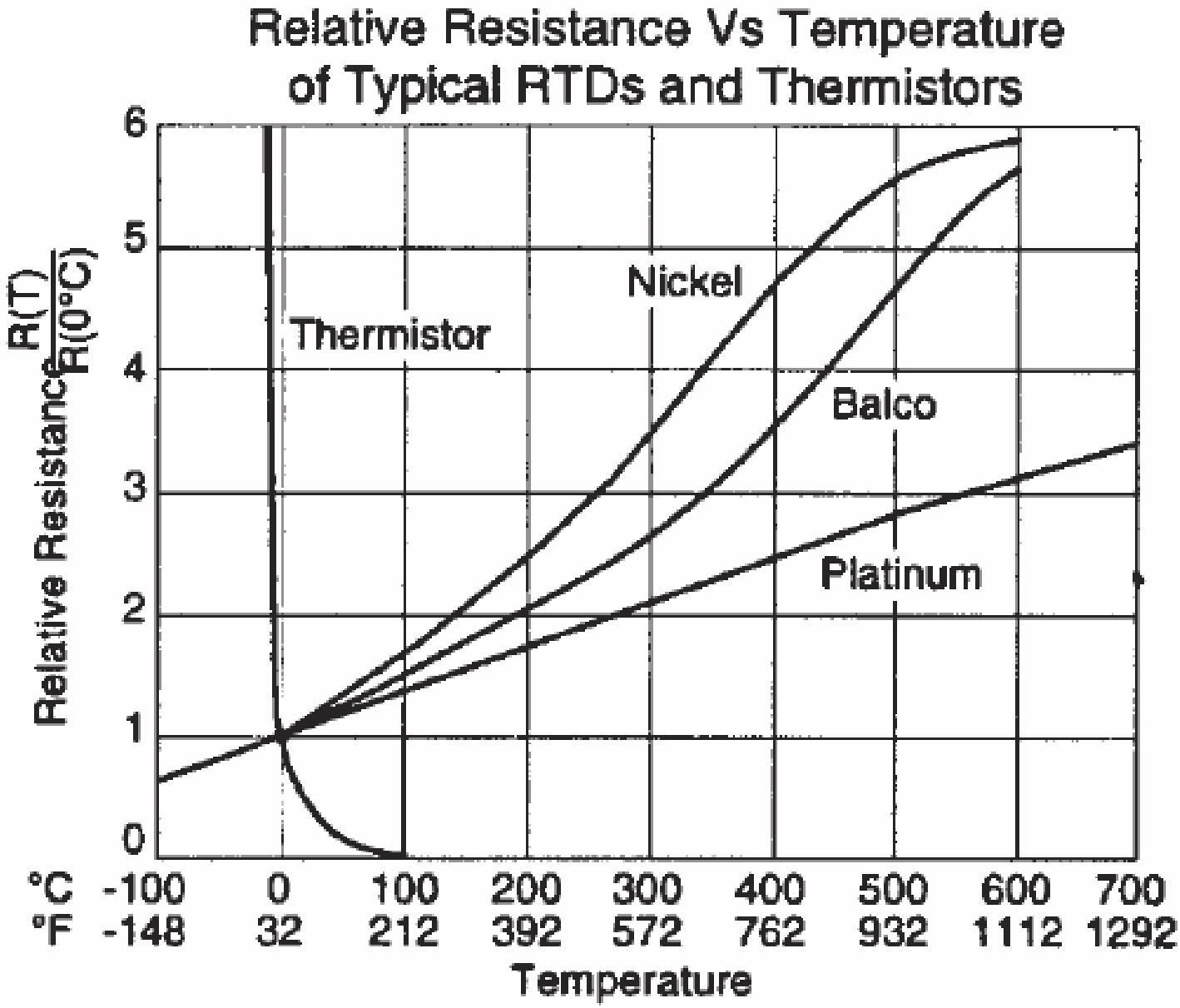



**Figure 7.7** *Schematic diagram of simple D.C. and A.C. resistance thermometry systems. In the A.C. system, the capacitors, C, isolate any D.C. component that might arise as a result of thermal e.m.f.s*


Relative Resistance Vs Temperature
of Typical RTDs and Thermistors
Relative Resistance R(T)/R(0°C)
6
5
4
3
2
1
0
Thermistor
Nickel
Balco
Platinum
°C -100 0 100 200 300 400 500 600 700
°F -148 32 212 392 572 762 932 1112 1292
Temperature

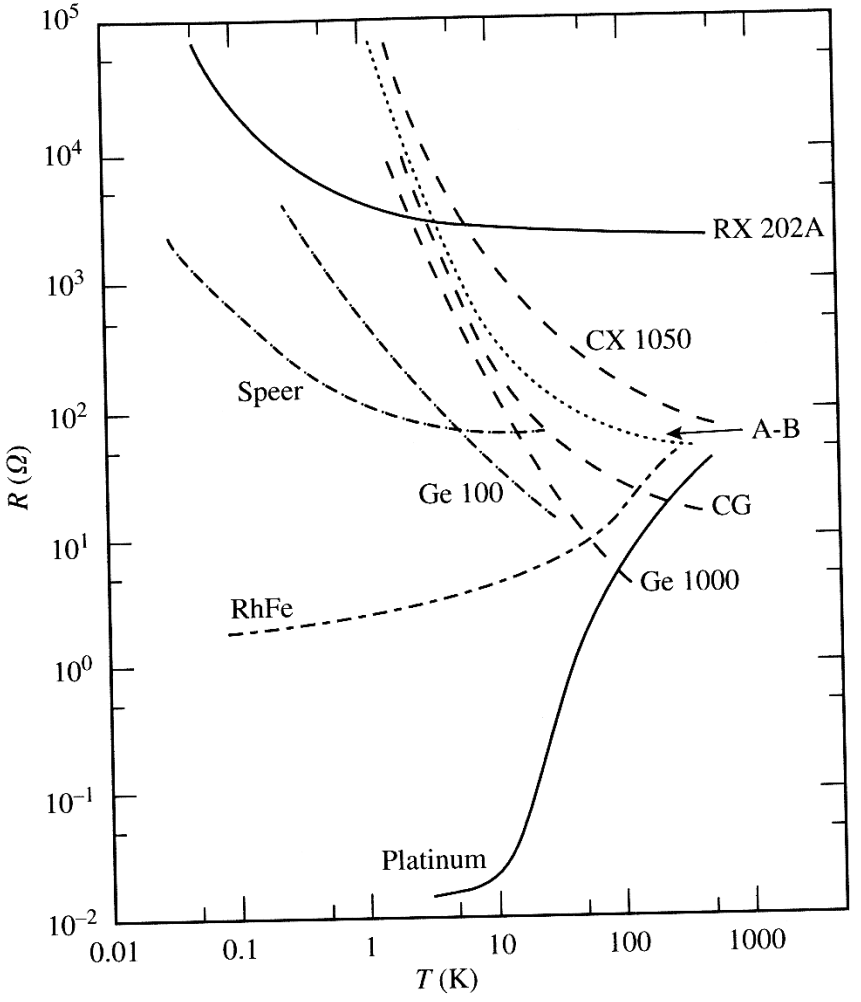



FIG. 3.7 The electrical resistance $R(T)$ of some typical thermometers. A-B denotes Allen-Bradley carbon resistor. Speer is a carbon resistor. CG is carbon-in-glass. CX 1050 is a Cernox and RX 202A is a ruthenium oxide from LakeShore. Ge 100 and Ge 1000 are Cryocal germanium thermometers.